

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **data pipelines** for operational technology systems, sourcing and integrating **network data from GIS, load/customer data from SAP, load profiles from RMDS, and weather data** using **Python, SQL, and ETL workflows** to enable advanced distribution management system (ADMS) capabilities.
- Executed **power flow studies** and load flow analysis to validate system accuracy against recorded measurements, identifying **25+ operational discrepancies** and implementing corrective actions that improved network reliability and system confidence scores by **18%**.
- Collaborated with **software providers** on data migration and system transitions, ensuring seamless handoffs of operational technology data from legacy systems (**Workbook, Apex**) to **Network Management Systems (NMS)** with **100% data integrity verification**.
- Analysed **large quantities of operational data** using **Excel, Oracle DB queries, and Power BI dashboards** to generate **KPI reports** and **load flow analytics** for cross-functional stakeholder review, supporting flexibility operations and network planning decisions.
- Managed production support and monitoring of enterprise systems using **Splunk, CloudWatch, and Dynatrace**, responding to **P1/P2 incidents** within SLA targets and implementing process improvements that reduced mean time to resolution (MTTR) by **22%**.
- Collaborated across **data analytics teams** and cross-functional business units to identify process improvement opportunities, document technical requirements, and transition pilot technologies and new propositions into operational delivery pipelines.

SKILLS

Power Systems & Network Modelling – power flow studies, load flow analysis, network data integration, system characteristics and parameters, GIS network data, RMDS load profiles.

Data Analytics & BI Tools – Power BI, Tableau, Excel, Oracle DB, SQL queries, data modelling, KPI reporting, large-scale data analysis.

Operational Technology Systems – Network Management Systems (NMS), ADMS (Advanced Distribution Management System), SAP (load/customer data), GIS, RMDS, Workbook, Apex, system integration.

Data Engineering & Pipelines – ETL design and implementation, Python scripting, data sourcing, data integration workflows, pipeline orchestration, schema management.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD pipelines, Jenkins, Git, Linux, cloud-native application monitoring.

Incident Management & Reliability – production support, system monitoring (Splunk, Dynatrace, CloudWatch), SLA management, P1/P2 incident response, mean time to resolution optimization.

Collaboration & Communication – cross-functional stakeholder engagement, technical documentation, presentation of analytics findings, requirements gathering, change management.

EDUCATION AND TRAINING

PROJECTS

Advanced Distribution Management System (ADMS) Data Integration & Validation (HCL Technologies)

- Architected end-to-end **data integration pipelines** to ingest **network data from GIS, load/customer data from SAP, microgeneration data, load profiles from RMDS, and weather data** into a centralized operational data warehouse, enabling unified visibility across distribution network operations.
- Conducted **power flow studies** and **load flow analysis** on 15+ critical network segments, comparing system output against recorded field measurements and identifying **32 accuracy gaps** that were resolved through data reconciliation and model refinement, improving system confidence by **21%**.
- Collaborated with **software providers** to orchestrate transition of operational technology data from legacy systems (**Workbook** and **Apex**) to **Network Management Systems (NMS)**, managing change control, UAT coordination, and post-implementation verification with **zero data loss incidents**.
- Built **Power BI dashboards** and **Oracle DB analytical queries** to deliver **KPI-driven operational reports** to network operations and flexibility teams, enabling real-time visibility into load flow performance, asset utilization, and network constraint areas.

Enterprise Systems Monitoring & Process Optimization (HCL Technologies)

- Delivered **production support and incident management** for mission-critical enterprise systems, utilizing **Splunk, Dynatrace, and CloudWatch** for real-time system health monitoring, P1/P2 incident triage, and **SLA compliance tracking** across operational environments.
- Identified and implemented **12 process improvements** through root-cause analysis of recurring incidents and system failures, reducing **mean time to resolution (MTTR)** by **22%** and improving overall system reliability by **19%**.
- Automated operational reporting and alerting workflows using **Python scripting, SQL data pipelines, and scheduled ETL jobs**, generating daily **KPI reports** and performance metrics for stakeholder review and decision-making, reducing manual reporting overhead by **8 hours per week**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)